Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/02/2008
PI Name:	McKinley, Gareth Ph.D.		
Project Title:	The Effects of Shear History on the Extensional Rheology of Complex Fluids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/18/2003	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-120 NOTE: Received NCE to 9/30/2007, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
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Task Description:	The extensional viscosity is a fundamental physical property of all non-Newtonian materials, and cannot be determined from simple viscometric shear flow experiments. Constitutive equations for viscoelastic fluids such as dilute polymer solutions predict large changes in the extensional viscosity as the elongation rate is increased; however, the validity of these theories has only recently been confirmed due to paucity of experimental data that was available in extensional flows. To date, most quantitative rheological measurements in shear-free flows have only been possible for highly elastic or "stiff" materials such as polymer melts which can easily be elongated without sagging under a gravitational body force. In a review for the National Research Council on the future of Fluid Dynamics Research, Denn (1993) remarked that "...there are two major outstanding problems in rheological measurements. One is the measurement of extensional stresses of mobile fluids such as polymer solutions." Previous research on filament stretching rheometry supported by NASA in the P.I.'s lab has gone at least some way to addressing this situation. The perturbative effects of gravity and/or container boundaries that have interfered with almost all previous studies of extensional rheology in mobile fluids have been addressed in two ways; (i) by miniaturizing the apparatus or using neutral buoyancy tanks to minimize body forces, or (ii) by performing similar experiments in an extended microgravity environment. Using the instrumentation developed it has become possible to obtain accurate measurements of the extensional viscosity for more 'mobile' fluids such as dilute and concentrated polymer solutions, gels, suspensions and liquid crystalline materials. This rheological data is important in validating or extending constitutive models that aim to describe the response of complex fluids in arbitrary deformation fields. Such models will ultimately allow designers of both space- and ground-based material processes to use numerical simulations for a priori predictions of the properties of new materials that involve complex two- and three-dimensional fluid flows. The objectives of the proposed research are to (i) adapt an existing filament stretching device developed at MIT in order to enable extensional measurements of a wider range of complex fluids including worm-like micellar fluids and associative polymer gels; (ii) To combine these mechanical measurements with a microstructural probe of the evolution of the molecular orientation of the polymer chains using flow-induced birefringence; (iii) to explore, for the first time, the effects of a pre-shear history on the transient extensional viscosity in a uniaxial stretching flow for dilute polymer solutions together with the subsequent relaxation behavior after cessation of extensional deformation; (iv) to investigate the viscoelastic free-surface adhesion instabilities and the process of fibril formation in viscoelastic fluids which contribute significantly to the work of adhesion that must be input to separate two rigid planar substrates. .
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge of the full rheological properties of complex fluids, including polymers, biological fluids, food materials, consumer products or, is necessary to accurately predict, model and improve their flow behavior. Understanding the extensional properties and the effects of shear history provides a scientific background for applications as diverse as industrial polymer processing, or the spinning of natural and synthetic spider silk.
Task Progress:	Improved versions of the filament stretching rheometer (FISER) instrument, developed previously in the PIs laboratory, have been built and specifically optimized to allow combined studies on both the shear history and the extensional behavior of complex fluids, in particular polymer solutions. The version developed at Glenn Research Center for the Microgravity Science Glovebox (MSG) combines both imaging and preshear capacities; this is the first instrument capable of accessing the effects of the shear history of a fluid on its extensional properties, enabled primarily by the extended range of time scales in the absence of gravitational sagging. The SHERE hardware has been launched aboard Space Shuttle Mission STS-120 to the International Space Station on October 23, 2007. In parallel, the instrumental sensitivity and the range of measurable stresses has been expanded, enabling greatly improved ground-based support for the ongoing MSG experiments aboard the ISS. We have performed a comprehensive rheological characterization of the model polymer solution ('Boger fluid') to be investigated in the MSG experiments. Furthermore, our instrumental improvements have made possible a much wider applicability of extensional rheometry to lower viscosity materials than were previously accessible. In consequence, a great number of additional classes of complex fluids can now be studied. Over the past year we have used our ground-based instrumentation to study systems as diverse as industrial adhesives, foods and consumer products, and biological materials.
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